

# Shenghan Gao

TELECOMMUNICATIONS · INTEGRATED PHOTONICS · COHERENT OPTICAL SYSTEMS

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## Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in coherent optical links, semiconductor-laser injection locking, and integrated photonic chip design. Dissertation work connects telecom-relevant carrier recovery, modulation-transfer analysis, fiber coherent-interference experiments, and chip-scale coherent processing nodes. Experienced with optical testbeds, lasers, EOMs, OSAs, SVAs, photodetectors, balanced detection, PID phase stabilization, KLayout photonic layouts, Lumerical/COMSOL/Zemax simulation, and Python/MATLAB data analysis. Targeting telecommunications, silicon photonics, coherent optics, and photonic hardware R&D roles.

## Core Skills

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|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Telecom optics</b>  | Coherent optical links, carrier recovery, optical injection locking, EOM modulation, sideband/RF analysis, balanced detection        |
| <b>Photonics chips</b> | KLayout, waveguides, grating couplers, splitters/combiners, interferometric paths, e-beam lithography, SEM, optical characterization |
| <b>Lab systems</b>     | Tunable lasers, slave/master laser setups, OSAs, SVAs, photodetectors, lock-in amplifiers, PID control, fiber stretchers, VOAs, PCs  |
| <b>Simulation</b>      | Semiconductor-laser rate equations, injection-ratio/detuning sweeps, modulation transfer, Lumerical, COMSOL, Zemax, CodeV            |
| <b>Software</b>        | Python, NumPy, SciPy, Pandas, Matplotlib, PyVISA/SCPI, NI MAX, MATLAB, C++, Git/GitHub, pytest                                       |
| <b>ML / CV</b>         | Machine learning coursework, PyTorch, TensorFlow, scikit-learn, OpenCV, CNN image recognition                                        |

## Dissertation and Research Experience

### University of Pittsburgh

Pittsburgh, PA

PH.D. RESEARCH ASSISTANT | ADVISOR: NATHAN YOUNGBLOOD

Spring 2020 - present

- Developed a dissertation framework for remote coherent optical processing using optical injection locking, targeting systems where remotely generated optical data must be processed without repeated O/E conversion, digitization, buffering, and re-transmission.
- Modeled semiconductor-laser injection locking with rate equations, sweeping injection ratio, detuning, linewidth-enhancement factor, and operating threshold to map stable carrier-locking regions using phase- and intensity-stability criteria.
- Analyzed AM/PM modulation transfer through injection-locked lasers, extracting AM-to-AM, AM-to-PM, PM-to-AM, and PM-to-PM response channels to identify operating regimes that recover a clean coherent carrier while suppressing unwanted remote sideband transfer.
- Built end-to-end coherent multiply-and-accumulate simulations using remote/local symbol sequences, interference, balanced photodetection, and temporal integration; showed that lower injection ratios can improve computational fidelity when locking margin remains sufficient.
- Designed and executed fiber injection-locking experiments with tunable master/slave lasers, VOAs, polarization control, OSAs, photodetectors, and SVA measurements; matched -15 dB experimental injection behavior to calibrated simulations using  $\alpha = 2$  and threshold 1.944.
- Implemented coherent fiber interference validation using an injection-locked local carrier, retained remote signal branch, lock-in detection, PID phase feedback, and fiber-stretcher control to stabilize phase-sensitive measurements.
- Designed chip-scale coherent-processing layouts with grating-coupler access, waveguide routing, splitting/combining structures, and interferometric paths to translate fiber-validated injection-locking workflows onto integrated photonic platforms.

### University of Rochester

Rochester, NY

RESEARCH ASSISTANT | ADVISOR: XI-CHENG ZHANG

Sep 2017 - Aug 2019

- Generated THz signals from liquid plasmas and analyzed how temperature, polarity, and viscosity affected emission strength and spectral content.
- Processed time- and frequency-domain experimental data with SNR, repeatability, sensitivity, and regression analysis; prototyped digital cloaking using lenslet arrays and MATLAB pixel remapping.

## Industry Experience

### Lumentum Operations LLC

San Jose, CA

RESEARCH INTERN | MENTOR: JOE WEN

May 2023 - Aug 2023

- Built MATLAB/Python workflows for FIR-filter equalization of communication signals distorted by intersymbol interference, connecting signal-processing design to recovered waveform quality.
- Collected measured signal data and applied FFT-based frequency analysis to compare filter responses, tune coefficients, and improve demodulated-signal recovery.

### Sunny Optics

Ningbo, China

RESEARCH INTERN | MENTOR: CHAO XU

May 2018 - Aug 2018

- Designed and optimized automotive LiDAR and mobile-camera lens systems in CodeV/Zemax against optical performance and packaging requirements.
- Worked with opto-mechanical engineers through lab integration, using test feedback to refine alignment and system functionality.

# Education

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## University of Pittsburgh

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING

Pittsburgh, PA  
Dec. 2026 expected

## Georgia Institute of Technology

M.S. IN COMPUTER SCIENCE, MACHINE LEARNING TRACK

Atlanta, GA  
May 2027 expected

## University of Rochester

M.S. IN OPTICS (THESIS TRACK)

Rochester, NY  
Aug. 2019

B.S. IN OPTICAL ENGINEERING

May 2017

B.S. IN APPLIED MATHEMATICS

May 2017

- Relevant training: optical communications, silicon photonics simulation, signal processing, software engineering, deep learning, circuit design, lens design, coating design, photonic device design, and computational imaging.

# Selected Publications and Talks

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## MOSAIC: A Multicore Optical Architecture for Scaling Up AI Acceleration

L. LIU, R. ZHU, **S. GAO**, X. XIN, Y. ZHANG, N. YOUNGBLOOD, J. YANG, SUBMITTED TO HPCA 2027, UNDER REVIEW, 2026

## Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

**GAO, S.**, K. LÜDGE, F. DA ROS, N. YOUNGBLOOD, SUBMITTED TO APL, UNDER REVIEW, 2026

## Distributed Coherent Photonic Computing via Injection Locking

**GAO, S.**, SPIE OPTICS + PHOTONICS (ORAL PRESENTATION), 2026

## Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

**GAO, S.**, INVITED PHD CANDIDATE TALK, PITTSBURGH QUANTUM INSTITUTE, PITTSBURGH, 2025

## Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

**GAO, S.**, P. R. SWINEHART, S. ZHONG, J. WUENSCHALL, N. LALAM, M. BURIC, K. P. CHEN, SPIE DEFENSE + COMMERCIAL SENSING (ORAL PRESENTATION), 2023

## Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. JIN, Y. E., **S. GAO**, X.-C. ZHANG, *Advanced Photonics* 2(1), 015001, 2020

## Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

**S. GAO**, Q. JIN, Y. E., X.-C. ZHANG, SPIE, 2019